Lab-5 Ex7 - Quadratic Equations Again

Find the discriminant and real number root(s) of a quadratic equation of the form:

$$ax^2 + bx + c = 0$$

Discriminant *D* is defined as:

$$\Delta = b^2 - 4ac$$

while **roots**, or **solutions**, are given by:

$$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad \text{and} \quad x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

https://en.wikipedia.org/wiki/Quadratic_formula

Specifically, your program should:

1. Read *three integers* `A`, `B` and `C`
2. Calculate the *discriminant* `D` and print it as the output result.
3. Determine number of **real** number solutions based on the discriminant value `D` and print the value of the **real number solution(s)**, i.e. *the x-intercept(s)* (if any), based on the following 3 cases:
   - `D > 0`: There are two real roots. Print out *both roots*.
   - `D = 0`: There is one real double root. Print out *the double root*.
   - `D < 0`: There are no real roots. Print out *the result*.

In this exercise, all the input values are assumed to be **non-zero integers**, while the output discriminant is of **integer** value and all the root value(s) shall have **two decimal places** only. You may exploit `<math.h>` in your work by inserting this at the start of your program:

```
#include <math.h>
```

which allows the use of the `sqrt()` function, for instance:

```
double x = sqrt(9.00);
```

If you are confused, please consult lecture notes "`03. Examples (Branching)`".

User-input numbers are ***bold and italic*** in the following sample runs.

Sample run #1:

```
Input A, B and C: 1 5 6
D = 1
The two roots are -2.00 and -3.00
```

Note: You must print the larger root first and then the smaller one (-2.00 > -3.00).

Sample run #2:

```
Input A, B and C: 1 -2 1
D = 0
The double root is 1.00
```

Sample run #3:

```
Input A, B and C: 5 4 2
D = -24
No real roots
```
